Homework 3

Collaboration with other students are encouraged; remember to name anyone you collaborate with and to write up your final solution by yourself.

When writing proofs, mathematical clarity is just as important as correctness and rigor. Please keep this in mind and try to adhere to proof writing conventions when typing up solutions.

1. Compute the remainder
$$7^{10^{1000}} \bmod 13$$
Hint: Use Fermat's Little Theorem.

2. A coin with probability $\frac{1}{3}$ of landing heads is tossed repeatedly until a head appears for the first time.
   (a) Compute the expected number of tosses using linearity of expectation.
   Hint: What is the probability that a head appears on first coin toss? What about second?
   (b) Compute the same expectation by deriving a recurrence relation for its expectation.

   Do not simply state the expectation of a geometric random variable without proof.

3. Let N be a list of integers and let $l, u \in \mathbb{Z}$. A contiguous sum $S(i, j)$ is defined by any $i \leq j$ such that
$$S(i, j) = \sum_{k=i}^{j} N[k].$$
   Provide an efficient divide and conquer algorithm that returns the number of contiguous sums in N that lie in the interval $[l, u]$. Justify the correctness and provide an analysis of your algorithm's runtime.

4. Let n balls be inserted sequentially into n bins. Initially, all bins are empty. At each time step $t = 1, 2, \ldots, n$, a new ball arrives and must be placed immediately into one of the bins according to a specified randomized allocation rule. Let $L_{\max}(n)$ denote the maximum load (i.e., the maximum number of balls in any bin) after all n balls have been placed.
   Provide an efficient randomized algorithm where the maximum load is upper bounded, with high probability, by:
$$L_{max}(n) = \mathcal{O}\left(\frac{\log n}{\log \log n}\right)$$
   Justify the correctness and provide an analysis of your algorithm's runtime.
   Hint1: One convenient form of the standard Chernoff-style inequality for binomial tails is:
$$\Pr[X \geq k] \leq \left(\frac{e\mu}{k}\right)^k \qquad \text{for } k \geq \mu.$$
   Hint2: An event E_n is said to occur *with high probability* (w.h.p.) if there exists a constant $c > 0$ such that
$$\Pr[E_n] \geq 1 - \frac{1}{n^c}$$
   for all sufficiently large n.